

- 4 The butterfly is an insect that feeds on nectar produced by flowers.

The photograph shows a butterfly feeding on a flower.



(Source: James Schwabel / Alamy Stock Photo)

- (a) The nectar in flowers contains nutrients including sugars, amino acids and lipids.

- (i) The sugars in the nectar are fructose, glucose and sucrose.

Which of these contain glycosidic bonds?

(1)

- ☐ **A** fructose only
- ☐ **B** sucrose only
- ☐ **C** fructose and glucose
- ☐ **D** fructose and sucrose



(ii) Which row of the table describes how amino acids are joined together to form a protein?

(1)

	name of bond	type of reaction
☐ A	ester	condensation
☐ B	ester	hydrolysis
☐ C	peptide	condensation
☐ D	peptide	hydrolysis

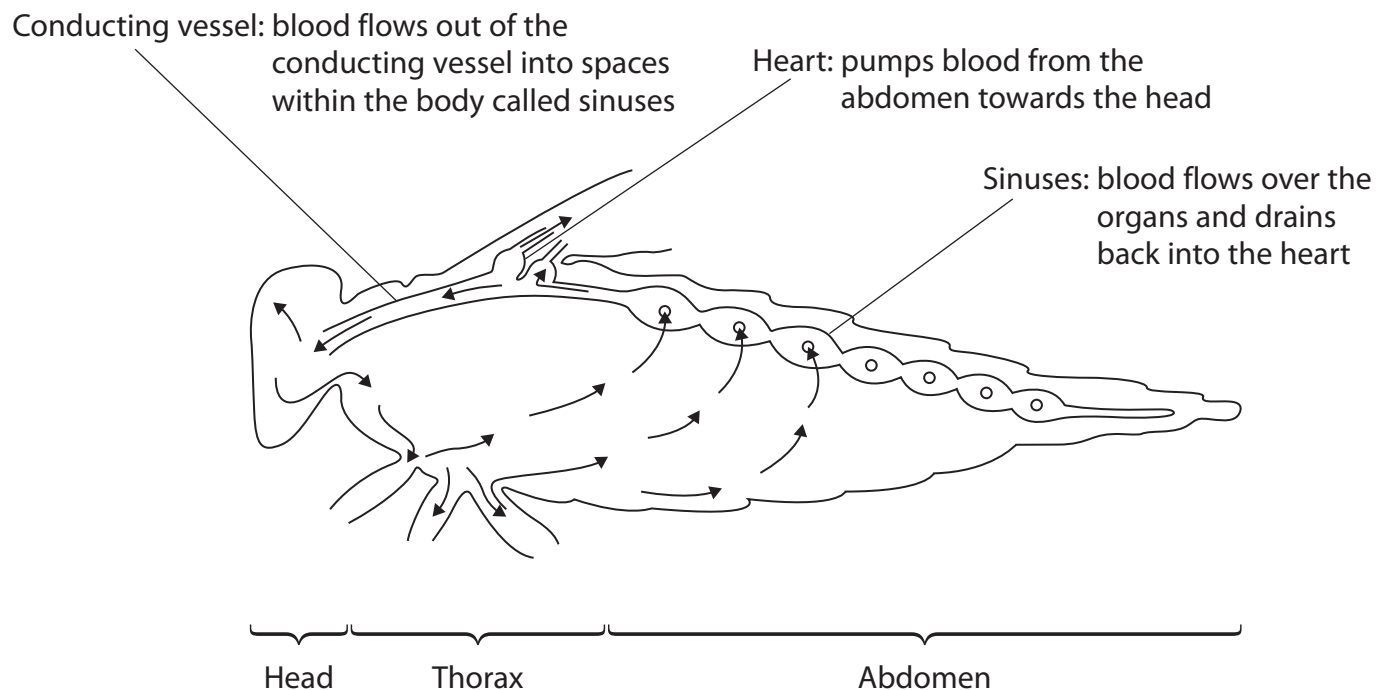
(iii) Which row of the table describes a saturated lipid?

(1)

	carbon-carbon double bonds	carbon : hydrogen ratio
☐ A	absent	higher than in an unsaturated fatty acid chain
☐ B	absent	lower than in an unsaturated fatty acid chain
☐ C	present	higher than in an unsaturated fatty acid chain
☐ D	present	lower than in an unsaturated fatty acid chain

- (b) The circulatory system of an insect is described as an open system. This means that the blood is not contained inside blood vessels but flows through cavities called sinuses.

The diagram shows part of the circulatory system of an insect.



- (i) The length of the head of a butterfly is 4 mm, the thorax is 6 mm and the abdomen is 18 mm.

Estimate the surface area to volume ratio of the butterfly.

Assume that the insect is a cylinder of diameter 4 mm and the surface area is 360 mm^2 .

(2)

Answer

- (ii) Explain why the circulation of a butterfly is different from the circulation of a mammal.

(2)

- (iii) The blood flowing through the sinuses of the butterfly is separated from the organs by collagen.

Describe the structure of collagen.

(2)

(Total for Question 4 = 9 marks)